Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 46

Finding Recurrences

Covers *MCS* Chapter 21

Status of this set

Ungraded practice. Solutions released with it.

Part I: How to Practice This Material

Setting up a recurrence

Ask: **what is the first choice, and what problem remains after it?**

- Disjoint first choices ⇒ **add** the resulting counts.
- Independent stages ⇒ **multiply**.
- Then state the boundary conditions — as many as the recurrence reaches back.

This is Lecture 31's counting with the answer written in terms of itself.

Three mistakes to expect

1. **Too few boundary conditions.** A recurrence reaching back two steps needs two.
2. **Overlapping cases.** The first choices must be mutually exclusive, or you double count.
3. **Treating a conjecture as proved.** Terms suggest; they do not establish.

Part II: Practice Problems

Section A — Writing the equation

1. A sequence satisfies $a_0 = 2$ and $a_n = 3a_{n-1}$.

(a) Compute a_0 through a_5.

(b) Conjecture a closed form.

(c) What would it take to prove your conjecture?

2. How many bit strings of length n contain no two consecutive 1s? Set up a recurrence with boundary conditions, then compute the first six values.

3. Into how many regions do n lines divide the plane, assuming no two are parallel and no three meet at a point? Set up a recurrence and find the first six values.

Section B — Reading a recurrence

4. For each, state how many boundary conditions are required and why.

(a) $f(n) = 5f(n-1)$

(b) $f(n) = f(n-1) + f(n-2)$

(c) $f(n) = f(n-1) + f(n-3)$

(d) $T(n) = 2T(n/2) + n$

Section C — Writing

5. In four to six complete sentences, explain why setting up a recurrence is a counting argument, using one of the problems above.